

Project Report

Name	Sampatakumar Venkatapur
College	Rajiv Gandhi Institute of Technology, Bengaluru
Program Name	Full Stack Web Development Portfolio

Project - 1

Name of the Project:

EduConnect - Student Collaboration Platform

Tools Used:

React 19, Firebase Auth SDK, Node.js, Express.js, Multer Middleware, MongoDB Atlas, Mongoose, Axios, Tailwind CSS, React-Toastify.

Working Procedure:

- Created a resource upload form capturing branch, scheme, subject code, tags, and document title.
- Integrated the HTML Form Data API to package local file attachments along with text metadata for transmission.
- Configured Multer middleware on the backend routes to process physical files and save them in a secure static uploads directory.
- Implemented resource records in MongoDB utilizing a Mongoose schema with status values defaulting to 'pending'.
- Built an administrative moderation portal where system admins review and moderate submitted resources.
- Programmed PUT and PATCH API routes to approve, reject, or modify uploads.
- Integrated physical file deletion routines to remove corresponding documents from disk storage when an admin deletes a record.
- Applied security checks on admin endpoints to restrict moderation privileges exclusively to verified administrator UIDs.
- Developed user-feedback indicators using React Toastify to notify admins of database modifications.
- Configured static directory serving in Express to allow verified users to search and download approved notes.

Learning Outcomes:

- Gained deep understanding of handling multipart form uploads containing text metadata and physical documents using Multer.
- Learned to configure static file hosting endpoints and file-system directory access in Express.js.
- Developed expertise in role-based access control (RBAC) to secure sensitive administrative endpoints.
- Mastered CRUD state transitions on Mongoose models for content moderation workflows.
- Discovered database cleaning techniques to ensure orphaned files are purged from disk storage during delete operations.

Project - 2

Name of the Project:

EduConnect - Real-Time Chat & Room Messaging System

Tools Used:

React 19, Socket.IO-Client, Firebase Auth SDK, Node.js, Express.js, Socket.IO Server, MongoDB Atlas, Mongoose.

Working Procedure:

- Programmed the main React chat interface to establish persistent WebSocket connections with an Express backend using the Socket.IO library.
- Configured socket handshake middleware on the backend to authenticate connections by decoding and validating Firebase JWT ID tokens.
- Developed server-side and client-side socket event handlers (joinRoom, leaveRoom, sendMessage) to support multi-room chat channels and manage room transitions.
- Implemented real-time typing indicators and presence indicators to build an interactive community environment.
- Created optimistic UI update logic in React, rendering sent messages immediately on the client and reconciling them with database records once server confirmations arrived.
- Structured message documents using a Mongoose schema to store chat histories persistently in MongoDB.
- Set up history-fetching routines on room entries, fetching the latest 200 messages sorted by submission time.
- Designed signup and login screens with Firebase email verification guards to secure platform entries.
- Integrated conditional navigation routes based on user validation states to protect chatrooms from unauthenticated web traffic.

Learning Outcomes:

- Mastered bi-directional client-server communication using WebSockets via Socket.IO.
- Understood token-based WebSocket authentication using Firebase ID tokens and custom Express middleware.
- Gained experience handling concurrent data flows and optimistic state rendering inside React.
- Developed proficiency in structuring document schemas in Mongoose for fast real-time reads and writes.
- Learned client-side lifecycle management to prevent memory leaks by detaching socket listeners inside cleanup hooks.

Project - 3

Name of the Project:

SkillBridge - AI Skill Gap & Career Recommendation System

Tools Used:

React.js, Express.js, MongoDB Atlas, Mongoose, OpenAI API / Google Gemini API, Sentence Transformers, Tailwind CSS, Chart.js.

Working Procedure:

- Designed a student assessment dashboard displaying skill scores, strength graphs, and recommended tasks.
- Built assessment testing pages supporting multiple evaluation modes, including MCQ tests and logical reasoning modules.
- Integrated the OpenAI API and Google Gemini API on the Express backend to process student skills and compare them with target career roles.
- Utilized Sentence Transformers to perform vector-based similarity analyses, identifying missing skill areas.
- Programmed an AI learning roadmap generator that provides weekly and monthly learning roadmaps in structured JSON formats.
- Integrated roadmap recommendations with learning references, including YouTube tutorials, online courses, and documentation.
- Built a project recommendation filter suggesting beginner, intermediate, and advanced projects tailored to student skill scores.
- Created Mongoose schemas for roadmaps, skills, questions, and test results to log roadmap histories.
- Programmed dynamic chart interfaces using Chart.js in React to map student progress and streak data.
- Implemented gamification components, award badges, and experience point (XP) triggers to encourage learning consistency.

Learning Outcomes:

- Gained experience integrating third-party AI APIs (OpenAI & Google Gemini) to generate structured roadmap responses.
- Understood vector embeddings and similarity computations using Sentence Transformers to identify skill gaps.
- Learned to structure complex nested database collections in Mongoose for personal roadmap logs.
- Developed proficiency in rendering analytical charts and progress graphics from dynamically parsed API payloads.
- Learned to build assessment rules and scoring calculations to evaluate student competencies.

Project - 4

Name of the Project:

Zomato Reels - Short-Form Food & Hotel Video Platform

Tools Used:

React.js, Tailwind CSS, Framer Motion, Node.js, Express.js, Multer-Storage-Cloudinary, Cloudinary API, MongoDB, Mongoose.

Working Procedure:

- Designed a mobile-first user interface featuring a vertical scrolling reels feed displaying short-form videos of dishes and hotel reviews.
- Implemented play/pause triggers and volume controls that activate automatically based on the user's viewport scroll intersection.
- Developed a video upload form allowing restaurants, hotels, and food bloggers to submit videos, captions, and location tags.
- Configured Multer-Storage-Cloudinary on the Express backend to handle video uploads, streaming raw chunks to Cloudinary.
- Linked video documents to a MongoDB collection using Mongoose schemas containing video public URLs, descriptions, and hotel references.
- Programmed interaction buttons for liking, sharing, and bookmarking food reels to build user engagement on the frontend.
- Built location filter buttons allowing users to view food reels filmed near their current city or search by specific hotels.
- Optimized video loading using lazy-loaded video elements and thumbnail placeholders to reduce initial page-load bandwidth usage.
- Handled responsive layouts using Tailwind CSS, fitting perfectly on mobile formats and desktop viewports.

Learning Outcomes:

- Learned to handle large media assets and configure cloud video streaming pipelines using Cloudinary.
- Understood intersection observer APIs to manage video play/pause events during infinite scrolling.
- Gained experience using Multer and cloud storage integrations to manage user-uploaded video streams.
- Learned to design mobile-first UI components for vertical, full-screen video feeds.
- Understood database optimization strategies to store video metadata and references efficiently.

Project - 5

Name of the Project:

Personal Developer Portfolio (sampatakumar.github.io)

Tools Used:

HTML5, CSS3, JavaScript (ES6+), React.js, Framer Motion, Tailwind CSS, GitHub Pages, Git.

Working Procedure:

- Built the personal portfolio website deployed at sampatakumar.github.io to showcase projects, skills, education, and achievements.
- Structured the user interface with sections: Hero, About Me, Skills, Projects (displaying EduConnect and SkillBridge), and Contact.
- Utilized Tailwind CSS to establish a modern typography system and custom glassmorphic styling across all page layouts.
- Implemented responsive grid layouts, styling mobile-first media queries to support smartphones, tablets, and wide monitors.
- Programmed page transitions and scroll animations using Framer Motion to create a fluid, premium visual presentation.
- Set up a dynamic project details modal showing project summaries, links to GitHub repositories, and live demo URLs.
- Integrated a contact form sending user messages to a personal email using third-party email routing APIs.
- Configured Git version control and set up automated GitHub Actions workflows to build and deploy the React website to GitHub Pages.
- Implemented a dark/light theme switch, storing user preferences in LocalStorage to maintain styling between page reloads.
- Optimized page load performance by compressing image assets and using lazy loading for project screenshot elements.

Learning Outcomes:

- Mastered client-side hosting configurations, configuring domain paths for GitHub Pages deployments.
- Learned advanced visual enhancement techniques using Framer Motion animations and Tailwind utility classes.
- Gained proficiency in responsive design practices, ensuring cross-browser styling compatibility.
- Understood version control workflows and automated CI/CD deployment pipelines using GitHub Actions.
- Learned to integrate client-side contact routing services without writing custom mail servers.